

Enhanced Image Inpainting using Transformer-GAN Hybrid Models

Group - 32

Presenters:

Veera Saiva Sai Lakshmi Kalyan (50595073)

Taran Mamidala (50604177)

Jupalli Sai Abhisekh Reddy (50604117)

CSE 676-B: Deep Learning

Instructor: **Dr. Alina Vereshchaka**, Spring 2025



Problem Statement

The objective of our Project image inpainting using GAN models is to realistically fill in any damaged or missing areas of an image. While traditional models like BIG-GAN perform well, they frequently fall short in maintaining fine details or achieving results that are smooth and realistic-looking.

Key Challenges:

Blur Images:

- GAN-based models may generate **blurry or inconsistent textures**, especially in large missing regions.

Loss & Poor Structural consistency:

- It's hard to maintain the original **shape, symmetry, or geometry** of objects in the image..
- Models may **lose fine textures**, making the inpainted regions look artificial or mismatched

Purpose:

To Build a **Transformer-GAN hybrid model** that combines:

- **Transformers** for capturing the **global structure** of the image
- **GANs** for generating **fine, detailed textures**

Background

Introduction to Image inpainting:

What is Image Inpainting?

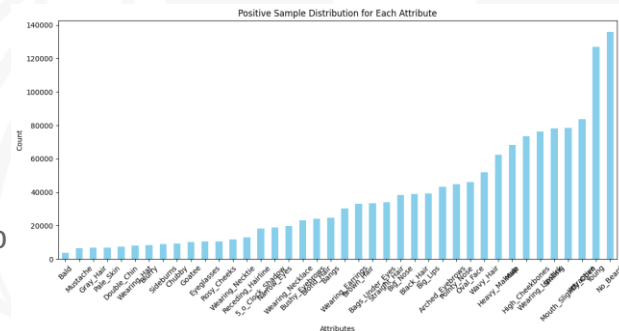
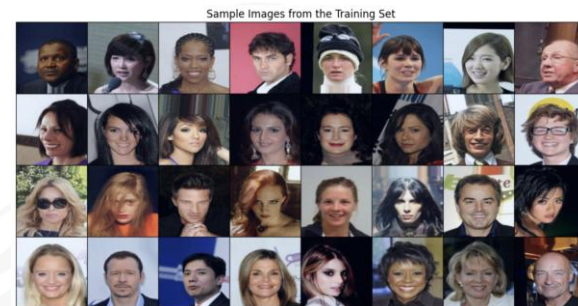
The process of restoring damaged or missing areas of an image is known as image inpainting. Making the finished image appear natural and fit in perfectly with the original information is its aim.

Importance:

- Useful in **photo restoration**.
- Helps in **removing unwanted objects** from images.
- Supports **medical imaging** by reconstructing missing or noisy data.
- Powers **AI-based creative tools**.

Real world Applications:

- Fixes old or damaged photographs by filling in missing or scratched parts so the image looks complete and fresh.
- Removes unwanted things from images (like people in the background or watermarks) and fills the gap naturally.
- Reconstructs missing parts in medical scans like MRIs or X-rays, helping doctors get a clearer view of organs or tissues.



Dataset Used:

Dataset:

Contains over **200,000** celebrity face images.

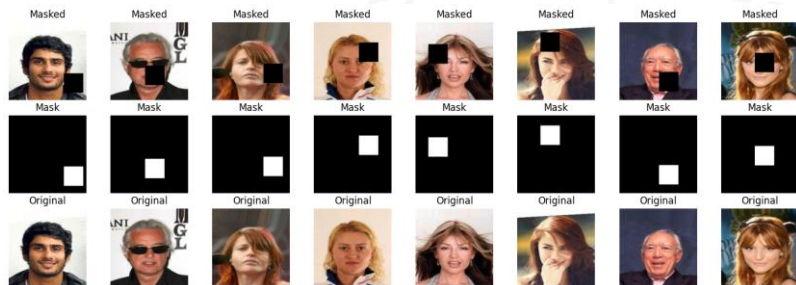
Each image is **cropped and resized** to 224×224 resolution.

With 40 binary attributes (e.g. Smiling, Eyeglasses)

Train: 162 770 samples, Val: 19 867 samples Test: 19 962 samples

Preprocessing Steps applied:

- **Image Normalization** to scale pixel values to $[-1, 1]$.
- Applied **random square masking and also freehand random masking** to simulate missing image regions (56×56).
- Generated **input pairs**: (masked image, original image).
- Used **PyTorch DataLoader** for efficient batching and shuffling.



Why?

We did these preprocessing steps to teach the model how to fill in missing parts of an image. By randomly masking parts of faces, the model learns to guess what should be there. Normalizing images helps the training go faster and stay stable. Creating consistent input-output pairs ensures the model can compare its guess with the correct answer.

GAN:

What is GAN?

A GAN is a type of deep learning model with two parts:

- **Generator:** Tries to create realistic images.
- **Discriminator:** Tries to tell the difference between real and fake images.
They compete with each other, and this "game" helps the generator get better over time.

Why GAN?:

- GANs are **great at generating realistic textures and fine details**, which is critical when filling in missing parts of images.
- They can **learn the global structure and local patterns** of faces, making the inpainted region blend naturally.
- Unlike traditional CNN's or autoencoders, GANs focus on realism—not just reconstruction accuracy.

How GAN works?

Random Input to Generator

The process starts with random input like usually either noise (random numbers) or a partially masked image. This is fed into the **Generator**, which tries to create a full, realistic image from it.

Generator Creates a Fake Image

The **Generator** is like an artist that's learning to draw realistic faces. At first, the output is not very good, but With feedback, it keeps improving. It produces a fake image that tries to look like a real image from the training data.

Discriminator Evaluates the Image

The **Discriminator** is like a judge. It looks at both real images (from the dataset) and fake images and tries to classify them as “real” or “fake.”

- If it identifies the fake correctly, it improves its judging skills.
- If it is fooled by the Generator, it updates its parameters to become more accurate.

Feedback for Both Models

Both models learn from this the Generator improves so it can better **fool** the Discriminator. The Discriminator improves so it can better **detect fakes**.

Method

We implemented and compared three different GAN-based architectures for image inpainting:

DCGAN-Based Generator with Mask-Augmentation

- Uses a **custom encoder-decoder structure** improved by explicit **mask processing** to guide inpainting.
- Mask and image features are encoded separately and fused before decoding.
- Losses: Binary Cross-Entropy (BCE) for adversarial loss + L1 pixel loss.

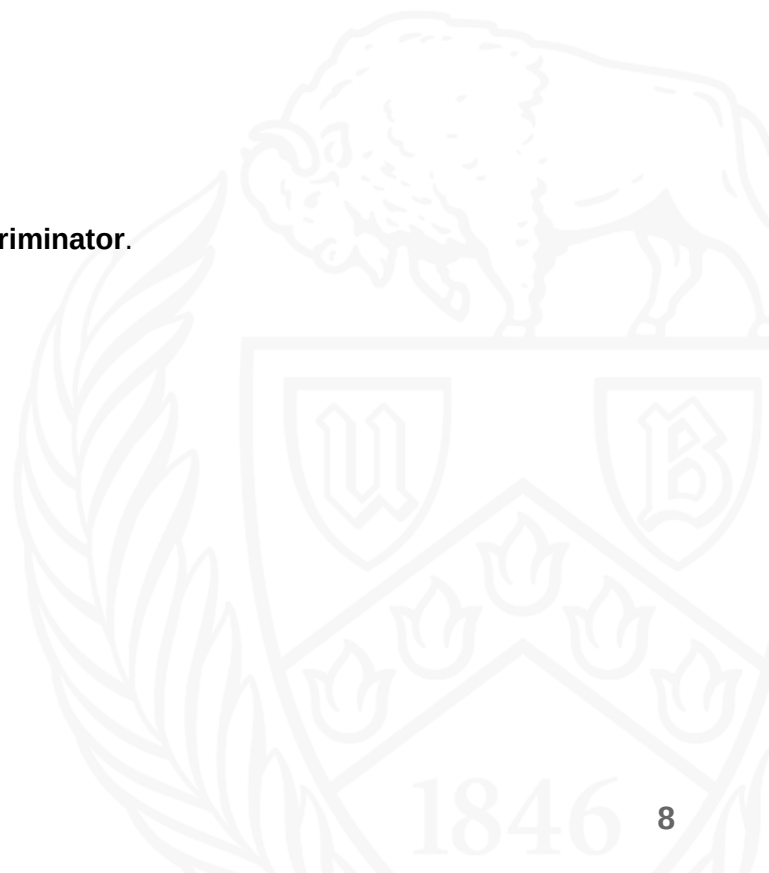
BigGAN-Inspired Residual Generator with Attention

- Employs a deeper encoder-decoder network with **residual blocks and attention layers** to retain global and local features.
- Aims to improve semantic consistency in large masked regions.
- Uses PatchGAN-style discriminator with **residual attention** for realism.

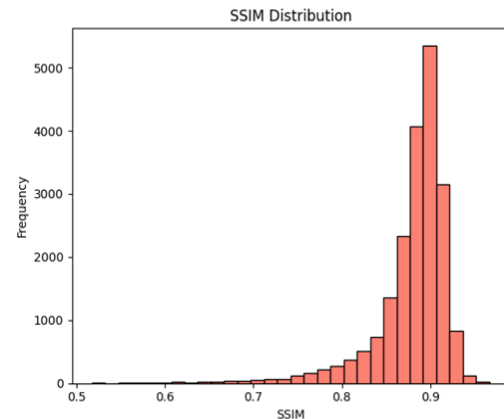
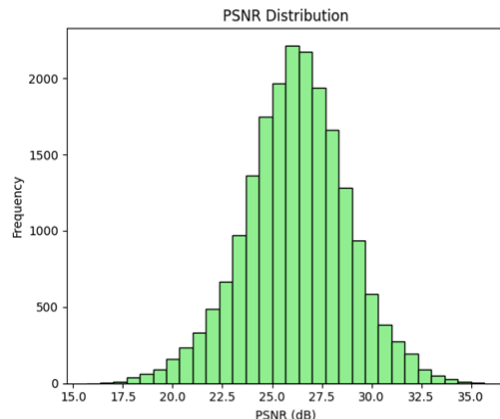
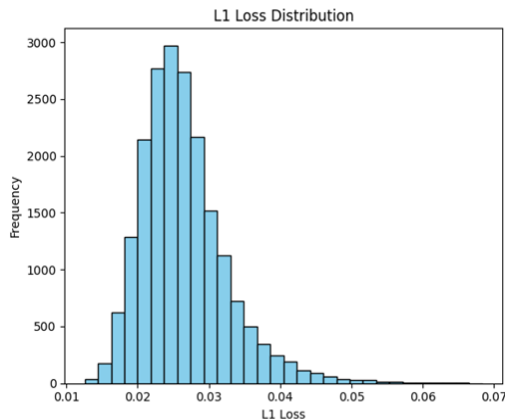
Method

Transformer-GAN Hybrid with Perceptual Loss

- Builds on a U-Net backbone with **residual blocks** and a **PatchGAN discriminator**.
- Optimized using a combination of:
 - **Adversarial loss** (BCE),
 - **L1 loss** for reconstruction,
 - **Perceptual loss** (VGG features),
 - **SSIM loss** for structural similarity.
- Focuses on improving **texture realism** and **structural integrity**.



Results



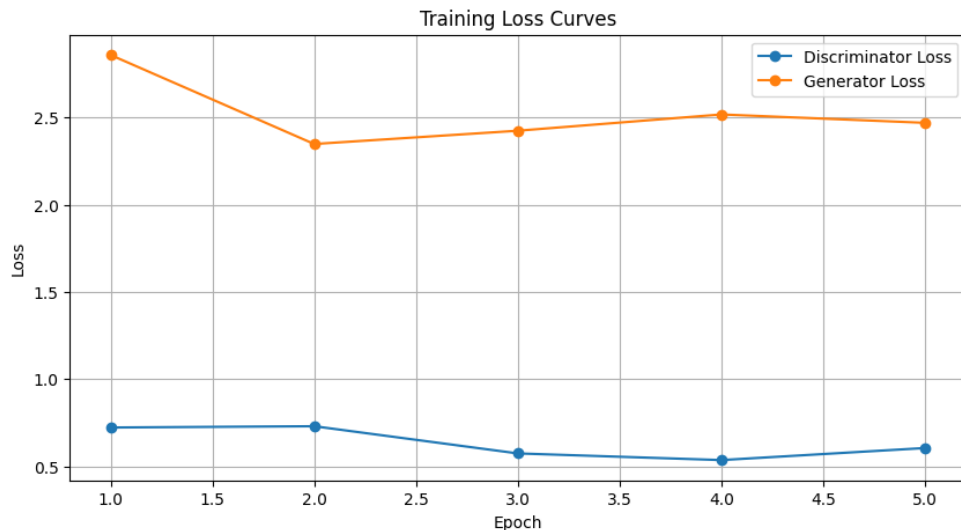
The L1 Loss distribution shows that the majority of inpainted images have very low pixel-wise error, with most values clustered around 0.025 to 0.035. This suggests that the generator is accurately restoring missing regions.

The PSNR distribution peaks around 26 dB, which indicates that the reconstructed images maintain good overall fidelity and clarity compared to the originals.

The SSIM distribution is skewed toward higher values (mostly above 0.85), showing that the model is successful in preserving structural and perceptual similarity, which is crucial for realistic inpainting.

Results

The training loss curve shows that the discriminator loss remains relatively stable across epochs, hovering around 0.6 to 0.7, which indicates balanced adversarial training. The generator loss initially decreases, suggesting early learning, but then slightly rises and stabilizes, which is typical in GANs as the generator tries to improve against a strengthening discriminator. Overall, both networks converge without signs of mode collapse or instability.



Observation

The Transformer-Gain Hybrid model received the best performance among all three models, with the lowest L1 loss and the highest PSNR and SSIM values. It was particularly effective in restoring large missing areas.

Adding attention mechanisms and residual connections, as seen in the Biggun-inspired model and transformer-song, improved the generation of fine texture and spatial continuity.

The use of advanced loss functions, including conceptual loss and SSIM loss, helps the model produce innapted images that are visually realistic and structurally accurate.

The transformer-gain model displayed that the global reference capabilities of the transformer by combining the GANS with the normal power of the GANS leads to more effective and natural image reconstruction.

Compared to DCGAN and Biggan models, Transformer-Gain Hybrid got better performance, with the SSIM score 0.88, which indicates a high structural similarity for the original unwanted images.

Key Observations/ Summary

- **Innovative Representation:**
 - Bipartite Graph model captures the entire structure of MILP nodes.
- **Siamese GNN Benefits:**
 - A strong node rankings learns function, which reduces the number of discovered nodes.
- **Performance Gains:**
 - Better than traditional SCIP heuristics and other ML models with the time of rapid solution.
- **Generalization:**
 - Effective on both training and large transfer examples.

Future Directions:

- End-to-end learned to integrate node selection with variable selection for Milp Solver
- Explore more advanced copy learning strategies or include reinforcement learning.
- **Real-World Impact:** - Potential benefits in industries such as logistics, energy, and finance. - Improved solver efficiency can lead to significant cost and time savings.

BigGAN-inspired Inpainted Outputs



Inpainted Images (Inference)



Contribution Summary

Team Member	Presentation Part	Contribution %
Taran Mamidala	Transformer-GAN Hybrid with Perceptual Loss	33.3%
Veera Saiva Sai Lakshmi Kalyan	BigGAN-Inspired Residual Generator with Attention	33.3%
Jupalli Sai Abhisekh Reddy	DCGAN-Based Generator with Mask- Augmentation	33.3%

THANK YOU

We look forward to your questions and
discussions

